MCU

File: mcu.kicad_sch

Power

File: power.kicad_sch

Outputs

File: outputs.kicad_sch

Inputs

File: inputs.kicad_sch

Connectors and CAN

File: connectors.kicad_sch

Wideband controller

File: wideband.kicad_sch

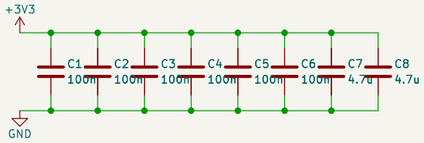
Miata ECU daughterboard rev 0.4 (on-board wideband controller, flashed through the main MCU). Block sheets connect through global labels; GND, +3V3, +5V and +12V are power symbols. J1 = power in and outputs (top ed

Sheet: /
File: miata_ecu_daughter.kicad_sch

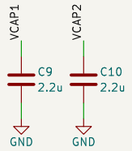
**Title: Miata ECU daughterboard**

| | | |
|---|---|---|
| Size: A4 | Date: 2026-10-29 | **Rev: 0.4** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 1/7 |

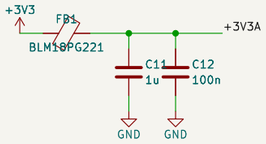
## Decoupling: one 100 nF per VDD pin, 2 x 4.7 uF bulk



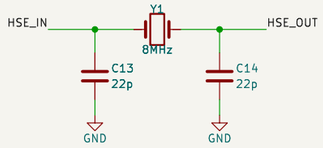

## Core regulator caps



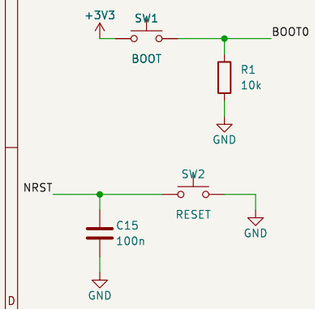

## Analog supply filter



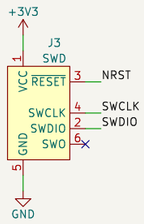

## 8 MHz HSE crystal

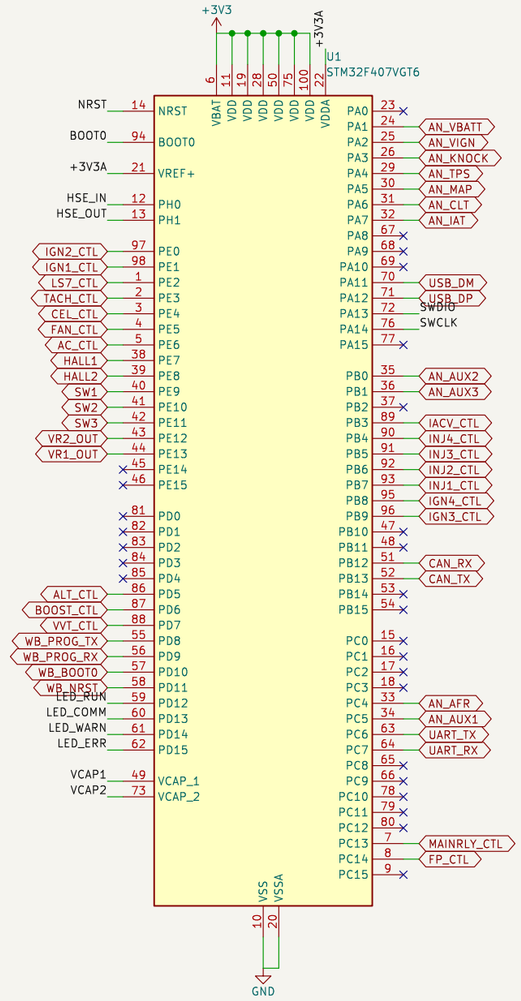



## BOOT0 (hold for DFU) and reset

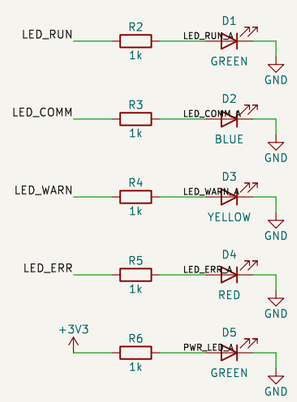



## SWD (Tag-Connect TC2030-NL pads)




## Status LEDs



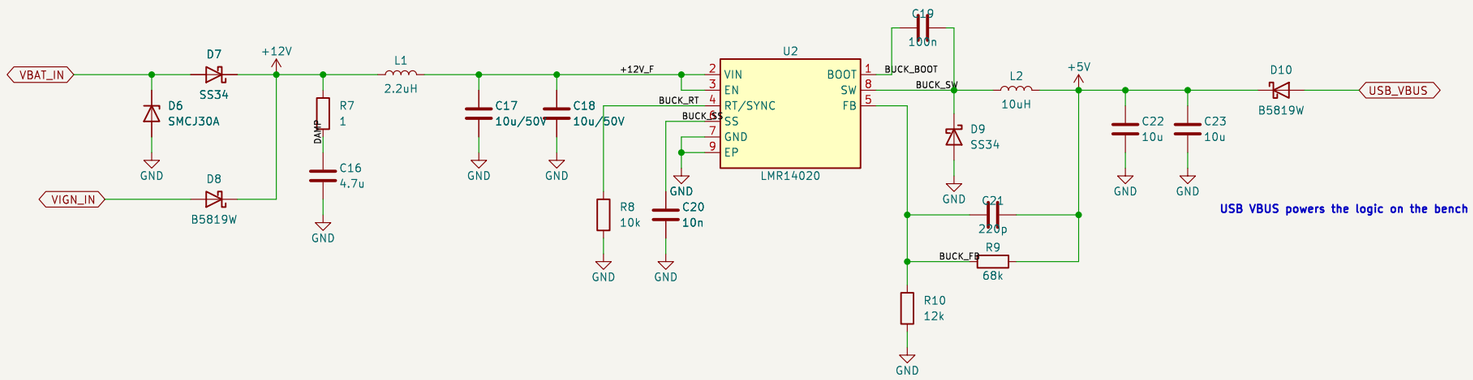

Generated from scripts/design.py; netlist identical to the routed board

Sheet: /MCU/
File: mcu.kicad_sch

**Title: Miata ECU daughterboard — MCU**

| Size: A3 | Date: 2026-10-29 | **Rev: 0.4** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 2/7 |

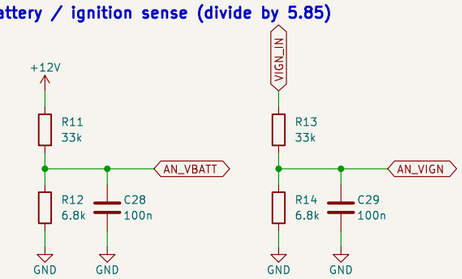
## Input protection and 5 V buck (LMR14020, 5 V / 2 A)



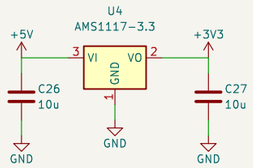

USB VBUS powers the logic on the bench

Fsw ~ 1 MHz (RT 10k); Vout = 0.75 V x (1 + 68k/12k) = 5.0 V

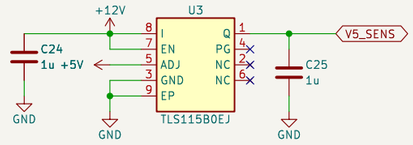
## Battery / ignition sense (divide by 5.85)




## 3.3 V LDO



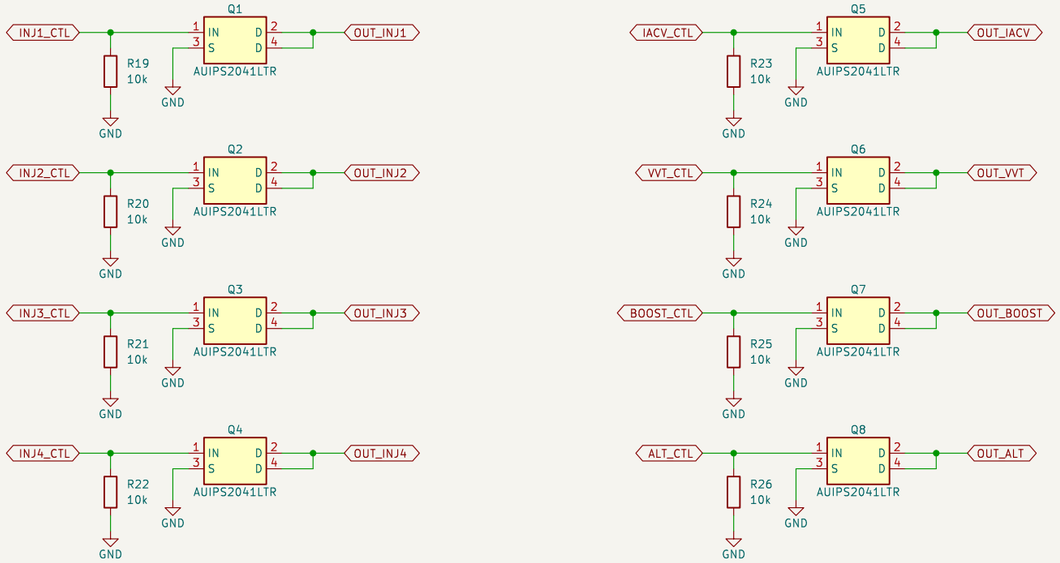

## Protected 5 V sensor supply (TLS115 tracks +5V)



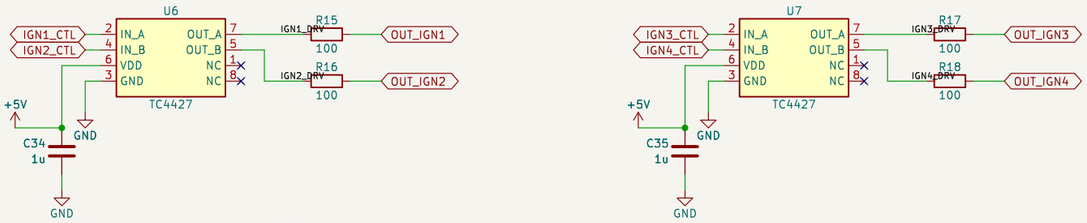

Generated from scripts/design.py; netlist identical to the routed board

| | | |
|---|---|---|
| Sheet: /Power/<br>File: power.kicad_sch | | |
| **Title: Miata ECU daughterboard — Power** | | |
| Size: A3 | Date: 2026-10-29 | **Rev: 0.4** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 3/7 |

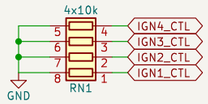
## Protected low-side drivers (AUIPS2041L): injectors, IACV, VVT, boost, alternator



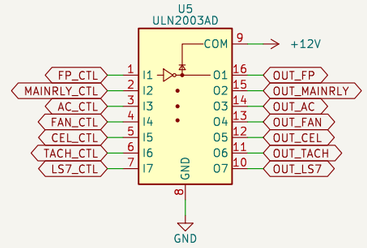

## Coil-on-plug logic outputs (TC4427, 5 V push-pull through 100 R)




Ignition inputs pulled low (coils stay off while the MCU boots)




## Relay, lamp and tach outputs (ULN2003, flyback to +12V)



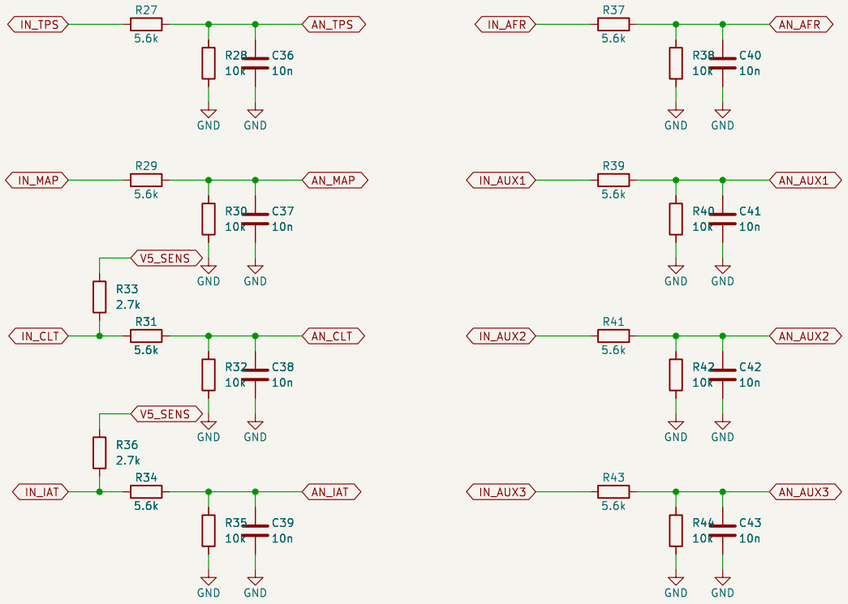

Generated from scripts/design.py; netlist identical to the routed board

Sheet: /Outputs/
File: outputs.kicad_sch

**Title: Miata ECU daughterboard — Outputs**

| Size: A3 | Date: 2026-10-29 | **Rev: 0.4** |
|---|---|---|
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 4/7 |

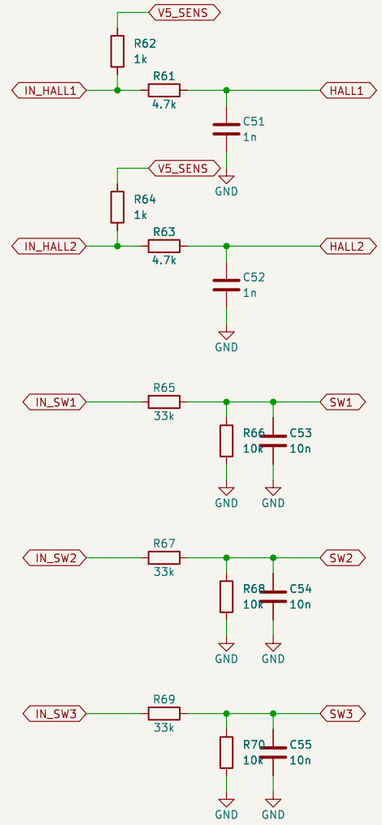
## Analog sensor inputs: 0–5 V -> 0–3.2 V for the ADC (5.6k / 10k, 10 nF)



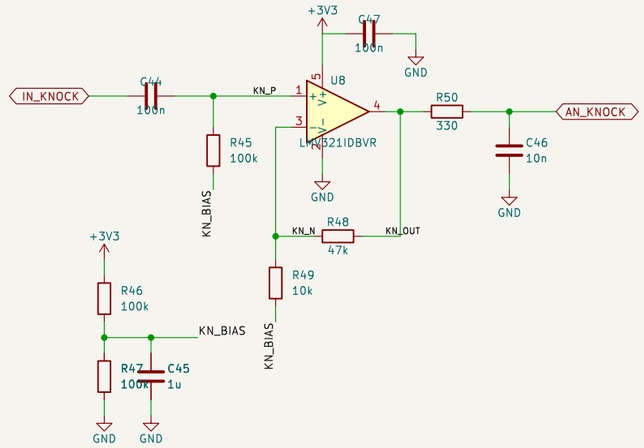




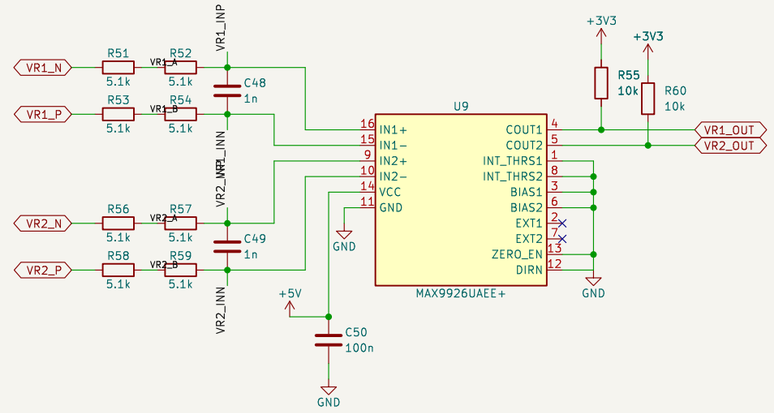
## Hall (open collector, 1k pull-up) and 12 V switch inputs




## Knock: AC-coupled, biased to 1.65 V, gain 5.7




## Crank / cam VR inputs: MAX9926 (outputs inverted, so harness + goes to IN−)



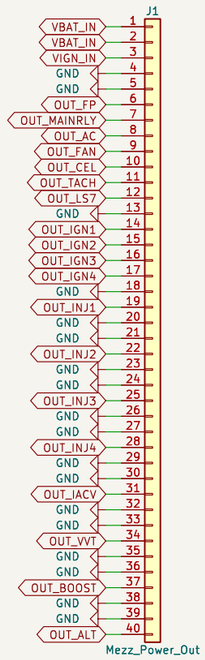

Generated from scripts/design.py; netlist identical to the routed board

Sheet: /Inputs/
File: inputs.kicad_sch

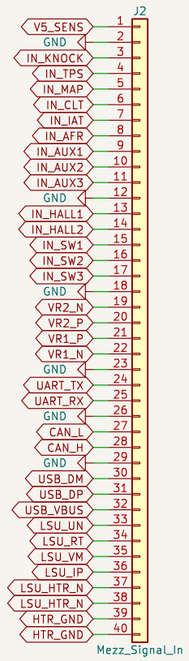
**Title: Miata ECU daughterboard — Inputs**

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-29 | **Rev: 0.4** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 5/7 |

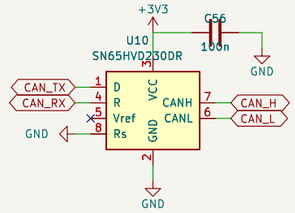
## J1: power in and outputs (top edge)




## J2: sensor inputs and comms (bottom edge)




## CAN transceiver (termination on the motherboard)




## Mounting holes




Generated from scripts/design.py; netlist identical to the routed board

Sheet: /Connectors and CAN/
File: connectors.kicad_sch

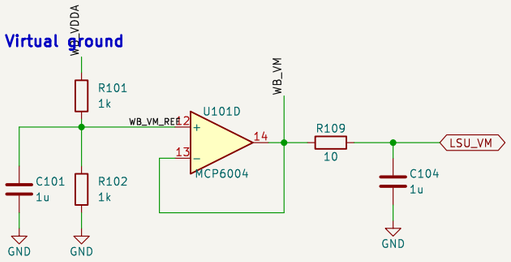
**Title: Miata ECU daughterboard — Connectors and CAN**

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-29 | **Rev: 0.4** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 6/7 |

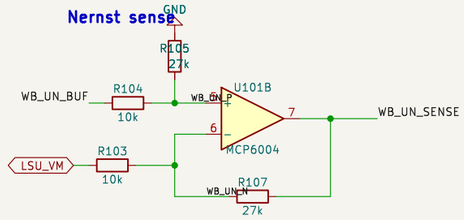
## Wideband O2 controller: FOME / rusEFI open-source wideband module (MIT), rev C circuit, laid out on this board

Refs are the module refs + 100 (module R31 = R131). Talks to the main MCU over the shared CAN bus.

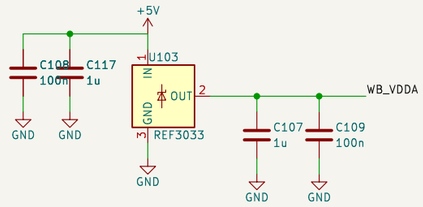
### Virtual ground



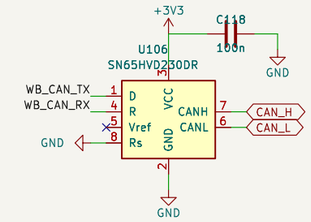

### Nernst sense



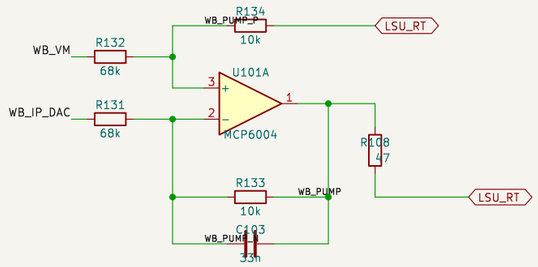

### ADC / analog reference



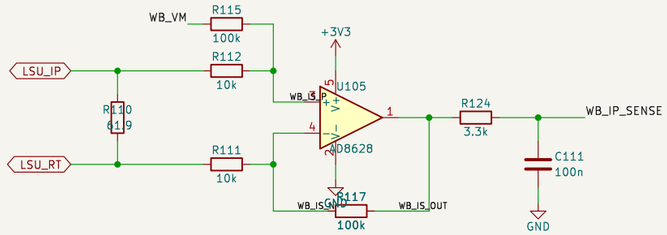

### CAN (same bus as the main MCU; rusEFI reads lambda over CAN)



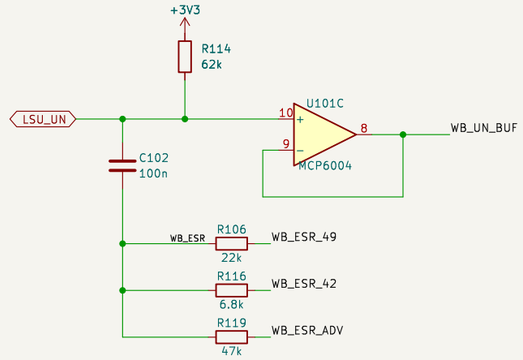

### Pump current driver (Ip DAC -> LSU pump cell)




### Pump current sense

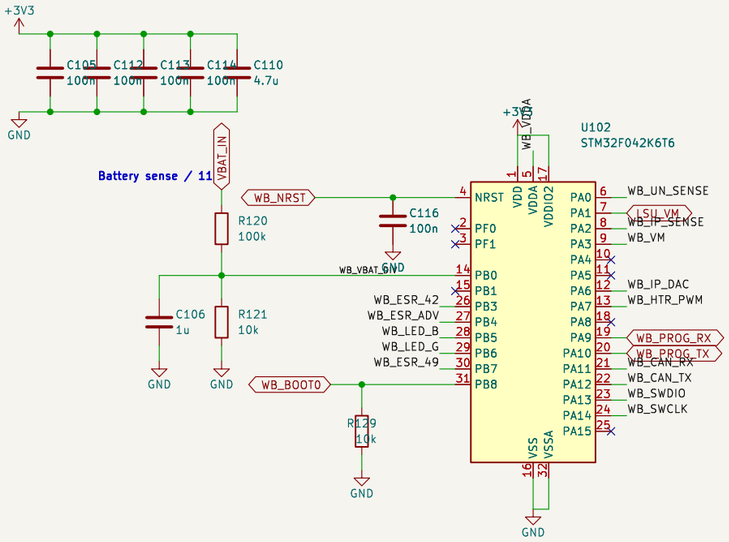



### Nernst buffer, bias current and cell resistance (ESR) drive

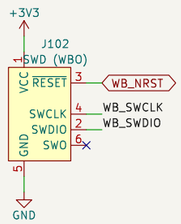



ESR drive: R106 22k / R116 6.8k / R119 47k, through C102 into the Nernst cell

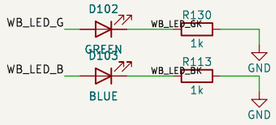
## Controller MCU (STM32F042, FOME wideband firmware; first flash over SWD, then over CAN)



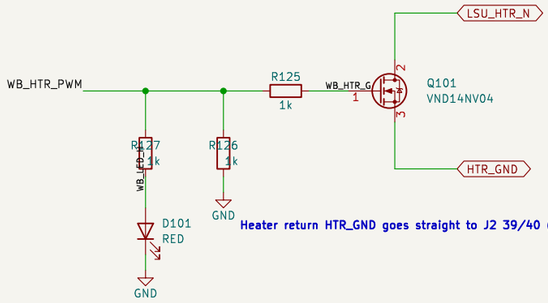

### SWD (Tag-Connect TC2030-NL pads, bottom side)




### Status LEDs




### LSU heater low side (protected OMNIFET)




Heater return HTR_GND goes straight to J2 39/40 (motherboard GND), not through this board's plane

### Op-amp supply




Generated from scripts/design.py; netlist identical to the routed board

| | | |
|---|---|---|
| Sheet: /Wideband controller/ | | |
| File: wideband.kicad_sch | | |
| **Title: Miata ECU daughterboard — Wideband controller** | | |
| Size: A3 | Date: 2026-10-29 | **Rev: 0.4** |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | | Id: 7/7 |